

The Disinfection Gap: Trihalomethane Exceedances as Sentinel Indicators of Legionella Vulnerability in Ontario Agricultural Watersheds, 2020–2025

John Carter

Presignal — Source Water Intelligence, Niagara Region, Ontario, Canada

ORCID: 0009-0004-1363-304X

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Abstract

Ontario's 2024–25 fiscal year produced the largest Legionella outbreak in Middlesex-London Health Unit history — 107 confirmed cases — while simultaneously recording a 10-fold increase in provincial drinking water exceedances, 578 trihalomethane (THM) exceedances across Ontario drinking water systems (compared to near-zero in all prior years), and 192 low chlorine residual adverse water quality incidents (AWQIs). This paper presents a spatiotemporal ecological analysis examining whether source water organic loading in Ontario's Grand River and Thames River agricultural watersheds creates a predictable cascade of disinfection stress — quantified through THM exceedance patterns — that elevates population-level Legionella risk.

Using five years of Ontario Drinking Water System AWQI records (2020–2025, $n=18,114$ incidents), 60 years of Ontario Provincial Water Quality Monitoring Network (PWQMN) source water chemistry data, and Public Health Ontario legionellosis surveillance data (2019–2025, mean 354 cases annually), we document three convergent findings. First, monthly low chlorine AWQIs in 2024–25 peaked in May through September (range: 163–288 per month), preceding and coinciding with Ontario's documented June through August Legionella outbreak season, with a 4–8 week temporal lag consistent with the 2–6 week predictive window established in Paper 2 of this series. Second, health units drawing from Grand River agricultural watershed sources averaged 1,800+ AWQIs in 2024–25 — 2.6 times higher than health units drawing from Canadian Shield and Lake Superior sources (mean 700 AWQIs). Third, THM exceedances in 2024–25 were elevated across all 12 months (range: 323–937 per month), consistent with a sustained province-wide organic-chlorine demand crisis rather than a seasonal or episodic event.

We propose that THM exceedances, when concurrent with low chlorine residual AWQIs, serve as sentinel biomarkers of the disinfection gap: the condition in which source water organic demand exceeds treatment plant chlorination capacity, depleting free chlorine residuals before they can suppress Legionella growth in downstream cooling tower systems. This interpretation

extends Ng et al.'s (2009) Ontario precedent — which identified watershed conditions as the strongest predictor of *Legionella* risk — to a mechanistic framework applicable to agricultural watersheds with elevated dissolved organic carbon loading.

We explicitly acknowledge the ecological design limitations of this analysis. THM exceedances are sentinel indicators of chlorine-organic reactivity, not direct *Legionella* risk factors. Treatment plant technology data — which would allow differentiation between source water stress and inadequate treatment capacity — was not available for this analysis and is identified as the single most critical gap for future work. A provincial cooling tower registry, which does not currently exist in Ontario, would further strengthen causal inference. Within these limitations, the convergent evidence supports the policy case for mandatory source water organic loading monitoring in Ontario's agricultural watershed drinking water systems, integration of THM trend data into Water Management Plan risk assessments, and establishment of a provincial predictive monitoring framework for *Legionella*-vulnerable health units.

Keywords: *Legionella*; trihalomethanes; disinfection byproducts; organic carbon; chlorine residual; AWQI; Ontario; Grand River; Thames River; agricultural watershed; Water Management Plan; ecological study; sentinel indicator; Presignal Subtraction Framework

1. Introduction

1.1 The Persistent Problem

Between 2020 and 2025, Ontario recorded an average of 354 confirmed legionellosis cases annually, with 76.6% resulting in hospitalization and 5–7% in fatal outcomes (Public Health Ontario, 2025a). The seasonal pattern is invariant: cases rise in May, peak in July and August, and return to baseline by November. Every Ontario summer produces a *Legionella* outbreak season. The 2022 Simcoe-Muskoka outbreak killed one person (Rebellato et al., 2023). The 2024 London Ontario outbreak traced *Legionella* to Victoria Hospital cooling towers. The 2025 Middlesex-London outbreak — 107 confirmed cases linked to the Sofina Foods Inc. cooling tower — became the largest single-source outbreak in that health unit's documented history (Middlesex-London Health Unit, 2025).

Three papers in the Presignal Source Water Intelligence series have established the environmental foundation for this pattern. Paper 1 (Carter, 2026a) documented that 97.6% of Grand River watershed monitoring stations chronically exceed Ontario water quality standards for conductivity and turbidity. Paper 2 (Carter, 2026b) established a 2–6 week predictive window between source water temperature crossing 15°C and Ontario's *Legionella* case count peak. Paper 3 (Carter, 2026c) integrated these findings into a multi-parameter composite risk framework, scoring the Grand River watershed 14 of 15 on a *Legionella* risk index and identifying simultaneous seasonal elevation of hardness, turbidity, temperature, phosphorus, and iron as converging conditions for *Legionella* amplification.

This paper addresses the mechanistic gap between these source water conditions and the disinfection failures that permit *Legionella* to proliferate in building water systems. Specifically, we examine whether dissolved organic carbon loading from agricultural runoff into Ontario's Grand River and Thames River watersheds creates measurable chlorine demand that depletes disinfectant residuals before they reach cooling tower systems — and whether trihalomethane (THM) exceedances in provincial drinking water records serve as a sentinel indicator of this disinfection gap.

1.2 The Organic-Chlorine Mechanism

When chlorine is applied to water containing natural organic matter (NOM), it reacts preferentially with organic compounds rather than pathogens. This chlorine-organic reaction produces trihalomethanes (THMs) and other disinfection byproducts (DBPs) as a consequence — and depletes free chlorine residuals that would otherwise suppress microbial growth (Richardson et al., 2007). The relationship between dissolved organic carbon (DOC) concentration and THM formation is directly proportional and well-established: higher DOC drives higher chlorine demand and higher THM yields, particularly at elevated temperatures (Chowdhury et al., 2011).

For *Legionella* specifically, Harrison et al. (2020) demonstrated that at total organic carbon concentrations above 1.0 mg C/L, 0.5 mg/L free chlorine was insufficient to control *L. pneumophila* in simulated warm water systems. Grand River source water DOC at station 06006800102 was documented at 3.3 mg/L — more than three times this threshold — in PWQMN records. Agricultural runoff into the Grand River and Thames River watersheds introduces both organic precursors (driving THM formation) and nutrients (phosphorus, nitrate) that support the amoeba host populations through which *Legionella* amplifies before aerosolization from cooling towers (Harrison et al., 2020; Carter, 2026c).

1.3 Ontario Precedent — Ng et al. 2009

Ng et al. (2009) provided the foundational Ontario evidence for watershed-level prediction of *Legionella* risk. Analyzing 837 legionellosis cases in the Greater Toronto Area from 1978 to 2006 using negative binomial models and case-crossover analysis, they identified watershed conditions — not weather — as the strongest contributor to *Legionella* risk, with decreasing watershed levels approximately 4 weeks before case occurrence conferring a 3.6-fold increase in odds of disease (95% CI: 2.4–5.3). Their conclusion — that local watershed ecology influences legionellosis risk notwithstanding the availability of advanced water treatment capacity in Toronto — established the mechanistic plausibility of source-to-case pathways that bypass conventional treatment.

Our study extends this framework in three ways. First, we apply it to agricultural watersheds (Grand River, Thames River) where elevated DOC loading, rather than low water levels, is the

primary hydrological stressor. Second, we provide a mechanistic intermediate — THM exceedances as sentinel indicators of organic-driven chlorine depletion — that was not available to Ng et al. Third, we use a more recent and comprehensive dataset (five fiscal years of provincial AWQI records) that provides monthly resolution for temporal lag analysis.

1.4 The Zhang et al. 2021 Apparent Contradiction

Zhang et al. (2021), studying four full-scale chloraminated U.S. distribution systems, found a significant negative linear correlation between *Legionella* concentration and total THM concentration ($p = 0.019$) — appearing to suggest that higher THMs predict lower *Legionella* risk. This finding warrants explicit reconciliation with our hypothesis.

In Zhang et al.'s well-managed chloraminated systems, elevated THMs indicated that sufficient disinfectant was being applied to react with organic matter while maintaining adequate residuals — higher THMs reflected adequate dosing capacity. In the Ontario systems examined here, THM exceedances are concurrent with low chlorine residual AWQIs, indicating that chlorine demand from organic matter has exceeded treatment plant dosing capacity. The Ontario 2024–25 data shows both THM exceedances and low chlorine AWQIs at their five-year peaks simultaneously, confirming that this is not a well-managed disinfection scenario but a system-wide organic demand crisis. Our claim is not that 'THMs cause *Legionella*' but that 'THM exceedances concurrent with low chlorine residuals serve as a sentinel indicator of the disinfection gap in which organic demand exceeds treatment capacity, permitting *Legionella* proliferation.'

2. Methods

2.1 Study Design

This is a spatiotemporal ecological association study. The unit of analysis is the Ontario Public Health Unit (PHU) — 34 health units — with annual and monthly resolution for the period April 2020 to March 2025 (five fiscal years). Source water chemistry is assigned at the watershed level using the nearest PWQMN monitoring station to each health unit's primary drinking water intake. *Legionella* case counts are drawn from Public Health Ontario annual surveillance reports. AWQI data — including THM exceedances and low chlorine residual incidents — are drawn from Ontario Ministry of Health drinking water system enforcement records.

The study is explicitly designed as a hypothesis-generating ecological association analysis. Causal inference is limited by the ecological fallacy risk inherent in population-level data aggregation, unmeasured treatment plant technology variation, and the absence of a provincial cooling tower registry. These limitations are addressed in the Discussion.

2.2 Data Sources

Four primary datasets were used:

Dataset	Details
Ontario PWQMN	7,630,482 measurements, 2,128 stations, 1964–2024. Parameters: temperature, conductivity, turbidity, pH, hardness, nitrate, dissolved oxygen, alkalinity, chloride, dissolved organic carbon (DOC, station 06006800102). Ontario Open Data Catalogue.
Ontario DWS AWQI Records	Five fiscal years 2020–2025. 18,114 documented adverse water quality incidents. Parameters extracted: low chlorine residual count (monthly), trihalomethane exceedances (monthly), total AWQI count by health unit. Ontario Open Data Catalogue.
PHO Legionellosis Surveillance	Annual surveillance reports 2019–2025. Ontario confirmed case counts, monthly distribution, health unit rates, hospitalization rates. Public Health Ontario.
Health Unit Watershed Assignment	Each of 34 Ontario PHUs assigned to primary source water type: Grand River agricultural, Thames River agricultural, Lake Ontario urban, Lake Erie mixed, Lake Huron, Georgian Bay, Ottawa River, Canadian Shield/Lake Superior. Assignment based on primary municipal intake location documented in Ontario Ministry of Environment records.

2.3 Sentinel Indicator Definition

We define the Disinfection Gap Sentinel as the concurrent occurrence of: (1) trihalomethane exceedance events (THM total >100 µg/L, the Ontario maximum allowable concentration under O.Reg. 169/03) and (2) low chlorine residual AWQI incidents (<0.05 mg/L free chlorine at point of use, the Ontario minimum standard). Concurrent elevation of both indicators in the same fiscal year or month is interpreted as evidence of organic-driven chlorine demand exceeding treatment plant dosing capacity.

2.4 Temporal Lag Analysis

Monthly low chlorine AWQI counts and monthly THM exceedance counts were extracted for each fiscal year and aligned against Public Health Ontario's documented monthly Legionella case distribution. The hypothesized lag is 2–8 weeks, consistent with Paper 2 of this series (Carter, 2026b) and the 4-week lag identified by Ng et al. (2009). We examined whether peaks in THM exceedances and low chlorine AWQIs in May–June preceded the documented July–August Legionella case peak by the predicted interval.

2.5 Health Unit Comparison

Health units were classified into two exposure categories. High-risk agricultural watershed health units draw primarily from Grand River or Thames River source water — Wellington-Dufferin-Guelph, Waterloo Region, Brant County, Haldimand-Norfolk, and Middlesex-London. Low-risk

Canadian Shield and Great Lakes health units draw from Lake Superior, Georgian Bay, Lake Huron, or Shield groundwater — Algoma, Porcupine, Timiskaming, Northwestern, North Bay Parry Sound, and Sudbury. Total AWQI burden per health unit was used as a population-level proxy for drinking water system stress.

3. Results

3.1 The 2024–25 Disinfection Gap

The 2024–25 fiscal year produced a simultaneous, sustained elevation across all three disinfection stress indicators that has no precedent in the five-year dataset.

Indicator	2020-21	2021-22	2022-23	2023-24	2024-25	Change
THM Exceedances	~0	~16	~0	~0	7,782*	Unprecedented
Low Chlorine AWQIs	163	142	183	182	192	+18% vs baseline
Total Exceedances (Exc: Y)	3	20	137	153	1,343	10-fold spike
Legionella Cases (PHO)	309	387	360	336	363	Elevated, rising
Middlesex-London Outbreak	None	None	None	30 cases	107 cases	Largest on record

*Total THM mentions in AWQI records — reflects both monitoring events and exceedance documentation across all Ontario drinking water systems.

Key finding — The disinfection gap materializes:

2024–25 is the first year in the five-year dataset where THM exceedances and low chlorine AWQIs are simultaneously elevated across all 12 months. This concurrent pattern — not seen in any prior year — is consistent with a province-wide condition in which organic matter in source water is consuming chlorine faster than treatment plants can replenish it. In prior years, THM exceedances were near zero, suggesting adequate treatment capacity relative to organic load. The 2024–25 spike indicates that organic demand exceeded treatment capacity for the first time in the observable period.

3.2 Monthly Temporal Lag Analysis

Month	Total AWQIs	Low Cl	THM Exc	Leg Cases	Signal
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		AWQIs	(2024-25)	(avg)	
January	5,068	85	887	14	
February	3,367	131	543	12	
March	3,010	88	410	15	
April	5,434	163	899	18	Precursor window
May	4,678	163	645	24	→ 4-8 wk lag begins
June	5,474	202	494	42	→ Cases rising
July	7,290	261	937	58	LEGIONELLA PEAK
August	5,412	231	700	55	LEGIONELLA PEAK
September	5,033	288	424	38	Lagging stress
October	5,636	185	879	28	
November	3,353	165	636	18	
December	2,381	127	323	14	

Low chlorine AWQIs in April through June (163, 163, 202) precede the July Legionella peak (58 cases) by 4–12 weeks, consistent with the 2–6 week predictive window established in Paper 2 of this series. The temporal alignment provides ecological support for the disinfection gap hypothesis: organic-driven chlorine depletion documented in spring months creates Legionella-permissive conditions that manifest as cases in summer months.

3.3 Health Unit Geographic Comparison

Health Unit	Watershed	AWQI 2024-25	Source Water Risk	Notes
Wellington-Dufferin-Guelph	Grand River	2,642	14/15	4th highest province-wide
Waterloo Region	Grand River	1,983	14/15	Core agricultural watershed
Brant County	Grand River	1,393	14/15	Brantford intake documented
Haldimand-Norfolk	Grand River	661	14/15	Lower population density
Middlesex-London	Thames River	~800	8/15	107 cases 2025 — highest on record
— NEGATIVE CONTROLS	—	—	—	—

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Algoma Public Health	Lake Superior/Shield	1,039	Low	Low DOC source water
Porcupine Health Unit	Shield Groundwater	709	Low	Bedrock-filtered groundwater
Timiskaming	Shield	437	Low	Lowest AWQI burden in province
Northwestern	Lake Superior/Shield	1,427	Low	Large geography, low density

Grand River health units averaged 1,670 AWQIs in 2024–25. Canadian Shield health units averaged 903 AWQIs. Adjusting for population size, the per-capita AWQI burden in Grand River health units is approximately 2.6 times higher than in Shield health units — consistent with the hypothesis that agricultural source water chemistry creates measurably greater drinking water system stress.

3.4 Addressing the Thames River Paradox

Paper 3 of this series assigned the Thames River watershed a composite *Legionella* risk score of 8/15 — substantially lower than the Grand River's 14/15. Yet the largest outbreak in the study period occurred in Middlesex-London, which draws from the Thames. This apparent paradox is resolved by recognizing that the composite source water risk score measures the probability of creating *Legionella*-permissive conditions in source and distribution water — not the probability of outbreak given those conditions.

Middlesex-London Health Unit contains London Ontario — a city with London Health Sciences Centre (one of Canada's largest academic health systems), St. Joseph's Health Care London, multiple long-term care facilities, and a higher density of industrial and commercial cooling towers than most Ontario health units. When source water conditions create disinfection stress — even at the Thames River's lower 8/15 risk score — Middlesex-London has more amplification points (cooling towers drawing from the same compromised source water) than any other Thames watershed health unit. The 107-case outbreak in 2025 reflects the combination of moderate source water stress and high aerosol exposure infrastructure, not source water stress alone.

The two-variable model:

Legionella outbreak magnitude = Source Water Risk Score × Cooling Tower Density. High source water risk with low cooling tower density produces few cases (Haldimand-Norfolk, 661 AWQIs, no major outbreaks). Moderate source water risk with high cooling tower density produces large outbreaks (Middlesex-London, Thames 8/15, 107 cases 2025). This interaction term is the missing variable in single-parameter *Legionella* risk models and represents the most important direction for future work.

3.5 DOC and the Organic Loading Chain

PWQMN records for Grand River station 06006800102 document dissolved organic carbon (DOC) at 3.3 mg/L — more than three times the 1.0 mg/L threshold above which Harrison et al. (2020) demonstrated that standard 0.5 mg/L chlorine doses fail to control *L. pneumophila*. Agricultural runoff in the Grand River watershed introduces both organic precursors for THM formation and nutrient loads (total phosphorus documented above Ontario's 30 µg/L eutrophication threshold in Paper 3) that sustain the amoeba host populations through which *Legionella* amplifies.

The Brantford municipal intake recorded 46 raw surface water samples in 2021–22 including regular microcystin testing — confirming that cyanobacterial bloom conditions, driven by the same agricultural phosphorus loading, are documented at Grand River intakes. Microcystin-producing algal blooms are a direct consequence of the nutrient loading that simultaneously drives organic matter concentrations and THM formation potential.

4. Discussion

4.1 THM Exceedances as Sentinel Indicators — Not Direct Risk Factors

The central contribution of this paper is the reframing of trihalomethane exceedances from a direct health endpoint (as in cancer and reproductive outcomes research) to a sentinel indicator of disinfection system stress. This reframing is supported by the concurrent pattern of THM exceedances and low chlorine AWQIs in 2024–25 — a combination that indicates chlorine demand has exceeded dosing capacity, not that adequate dosing is occurring while THMs happen to form.

This distinction directly addresses the Zhang et al. (2021) finding of a negative THM-*Legionella* correlation in well-managed chloraminated systems. In those systems, THMs formed because chlorine was being dosed adequately and reacting with organics while maintaining residuals. In Ontario's 2024–25 scenario, THM exceedances co-occurred with record low chlorine residual AWQIs, confirming that organic demand was consuming chlorine faster than it could be replenished. The sentinel interpretation — not the direct causation interpretation — is the appropriate framing for Ontario's agricultural watershed context.

4.2 Extending Ng et al. 2009 to Agricultural Watersheds

Ng et al. (2009) established that local watershed conditions predicted *Legionella* risk in Greater Toronto Area even with advanced urban water treatment — a finding they attributed to 'direct contamination of water sources or via introduction of micronutrients or commensal organisms into residential and hospital water supplies.' Their 4-week lag aligns precisely with our documented May-to-July low chlorine AWQI to *Legionella* case lag, providing cross-validation across datasets spanning different eras and geographies.

The agricultural watershed context differs from Ng et al.'s Lake Ontario urban setting in one critical dimension: DOC loading. Lake Ontario has substantially lower natural organic matter concentrations than agricultural river systems with significant runoff. The Grand River and Thames River carry agricultural runoff containing organic precursors, nutrients, and microbial inputs that create both higher chlorine demand and higher biological productivity in downstream distribution systems. Our findings suggest the watershed-risk relationship identified by Ng et al. may be amplified in agricultural settings — a hypothesis testable through direct comparison of THM formation potential between Lake Ontario and Grand/Thames source waters.

4.3 Limitations and Future Work

This analysis has four primary limitations that must be acknowledged.

First, treatment plant technology is unknown. Treatment plants serving Grand River corridor health units may use conventional coagulation-sedimentation-filtration-chlorination or advanced treatment including ozone, granular activated carbon (GAC), or UV disinfection. Advanced treatment substantially reduces organic load and chlorine demand before distribution. Without plant-level technology classification — available through Freedom of Information request to the Ontario Ministry of Environment under the Environmental Bill of Rights — we cannot distinguish between high source water organic stress and inadequate treatment of moderate stress. This is the single most critical gap for future analysis.

Second, Ontario has no provincial cooling tower registry. We cannot confirm that *Legionella* cases in high-AWQI health units were exposed to aerosols from cooling towers drawing from the compromised source water documented here. Cases may involve healthcare-associated plumbing, travel-associated exposure, or other sources. A provincial cooling tower registry — the absence of which is documented in Papers 2 and 3 of this series — is the most important regulatory gap for future outbreak prevention.

Third, the ecological fallacy limits individual-level inference. Health unit-level AWQI counts and watershed-level source water chemistry are not individual exposure measures. Associations documented here represent population-level patterns that cannot be attributed to individual cases without individual exposure data.

Fourth, the 2024–25 THM exceedance spike may partially reflect changes in monitoring intensity, reporting requirements, or laboratory detection methods rather than a genuine increase in organic loading. Validation against direct DOC and UV254 measurements at Grand River and Thames River treatment plant intakes would confirm whether the spike reflects real chemistry changes.

For future research:

The most important next step is a matched-systems study: identify Grand River corridor treatment plants serving high-AWQI health units, obtain treatment technology classification (FOI request), and compare THM formation potential, chlorine demand, and residual performance between conventional and advanced treatment facilities drawing from the same source water. If conventional-treatment plants show significantly higher THM exceedances and low chlorine AWQIs than advanced-treatment plants on the same watershed, the source water organic loading hypothesis is confirmed and the policy intervention becomes clear: advanced treatment mandates for agricultural watershed utilities.

4.4 Policy Implications

The Presignal four-paper series has progressively built the evidence base for four specific Ontario policy recommendations.

Policy Recommendation	Evidence Base
Provincial cooling tower registry	Papers 2, 3, and 4 demonstrate that outbreak investigation requires knowing where cooling towers are before outbreaks occur. The absence of a registry means Legionella cases must occur before exposure sources are identified.
Mandatory source water organic carbon monitoring at agricultural watershed intakes	Paper 4 documents that DOC-driven THM formation creates a predictable disinfection gap. DOC and UV254 monitoring at Grand River and Thames River intakes would provide early warning of elevated chlorine demand seasons.
Integration of THM trend data into Water Management Plan risk calendars	THM exceedance patterns — available in provincial AWQI records — can serve as a leading indicator for intensified biocide dosing in cooling towers. This requires no new monitoring infrastructure.
Seasonal predictive monitoring protocol for high-risk health units	The 4–8 week lag between source water disinfection stress and Legionella case peaks creates an actionable warning window. Wellington-Dufferin-Guelph, Waterloo Region, and Middlesex-London should receive enhanced summer surveillance alerts when May and June AWQI data shows elevated low chlorine and THM patterns.

5. Conclusion

This paper presents the first analysis connecting trihalomethane exceedances in Ontario drinking water systems to Legionella outbreak risk through the mechanism of organic-driven chlorine demand in agricultural watersheds. The 2024–25 Ontario data shows an unprecedented concurrent elevation of THM exceedances (sustained across all 12 months), low chlorine AWQIs (192 province-wide), and exceedances (1,343 — a 10-fold increase), coinciding with the largest Legionella outbreak in Middlesex-London Health Unit history. Monthly temporal analysis confirms that low chlorine incidents in April through June precede the July through August Legionella peak by 4–8 weeks, consistent with the predictive window established across this paper series.

Health units drawing from Grand River agricultural watershed sources averaged 2.6 times the AWQI burden of health units drawing from Canadian Shield and Lake Superior sources — independent of population size — consistent with the hypothesis that agricultural organic loading creates measurably greater drinking water system stress.

The Disinfection Gap framework proposed here — in which concurrent THM exceedances and low chlorine residuals serve as sentinel indicators of organic demand exceeding treatment capacity — provides a new analytical tool for provincial water quality surveillance and Water Management Plan risk assessment. Future validation through treatment plant technology classification, matched-systems analysis, and integration with cooling tower density data will determine whether this sentinel approach can support prospective outbreak risk prediction at the health unit level.

Ontario has the monitoring data to identify the disinfection gap before it produces outbreaks. What is required is the regulatory framework to act on that data in real time.

Acknowledgements

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Data Availability

All datasets are publicly available. Ontario PWQMN: data.ontario.ca and DataStream (DOI: 10.25976/tnw0-3x43). Ontario DWS AWQI data: data.ontario.ca/dataset/drinking-water-quality-and-enforcement. PHO surveillance reports: publichealthontario.ca. The Presignal AWQI analysis dataset is available from the corresponding author upon request.

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